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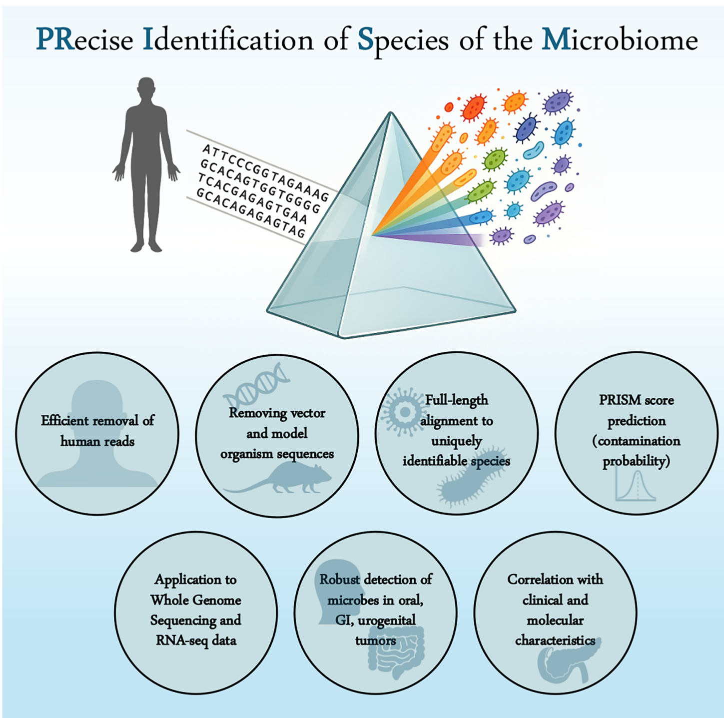
Report

Reliable detection of Host-Microbe Signatures in cancer using PRISM

Graphical abstract



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In brief

Ghaddar et al. develop and benchmark PRISM, a computational framework for precise microorganism identification and decontamination from human genomic data. Applying PRISM to TCGA and CPTAC, they uncover robust microbial signatures in selected tumor types and link microbial presence to tumor molecular and clinical features.

Highlights

- PRISM enables precise identification of microbial signals from human genomic data
- Oral, GI, and urogenital tumors in TCGA and CPTAC have robustly detectable microbes
- Microbes in pancreatic cancer in CPTAC associate with altered glycosylation

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Reliable detection of Host-Microbe Signatures in cancer using PRISM

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SUMMARY

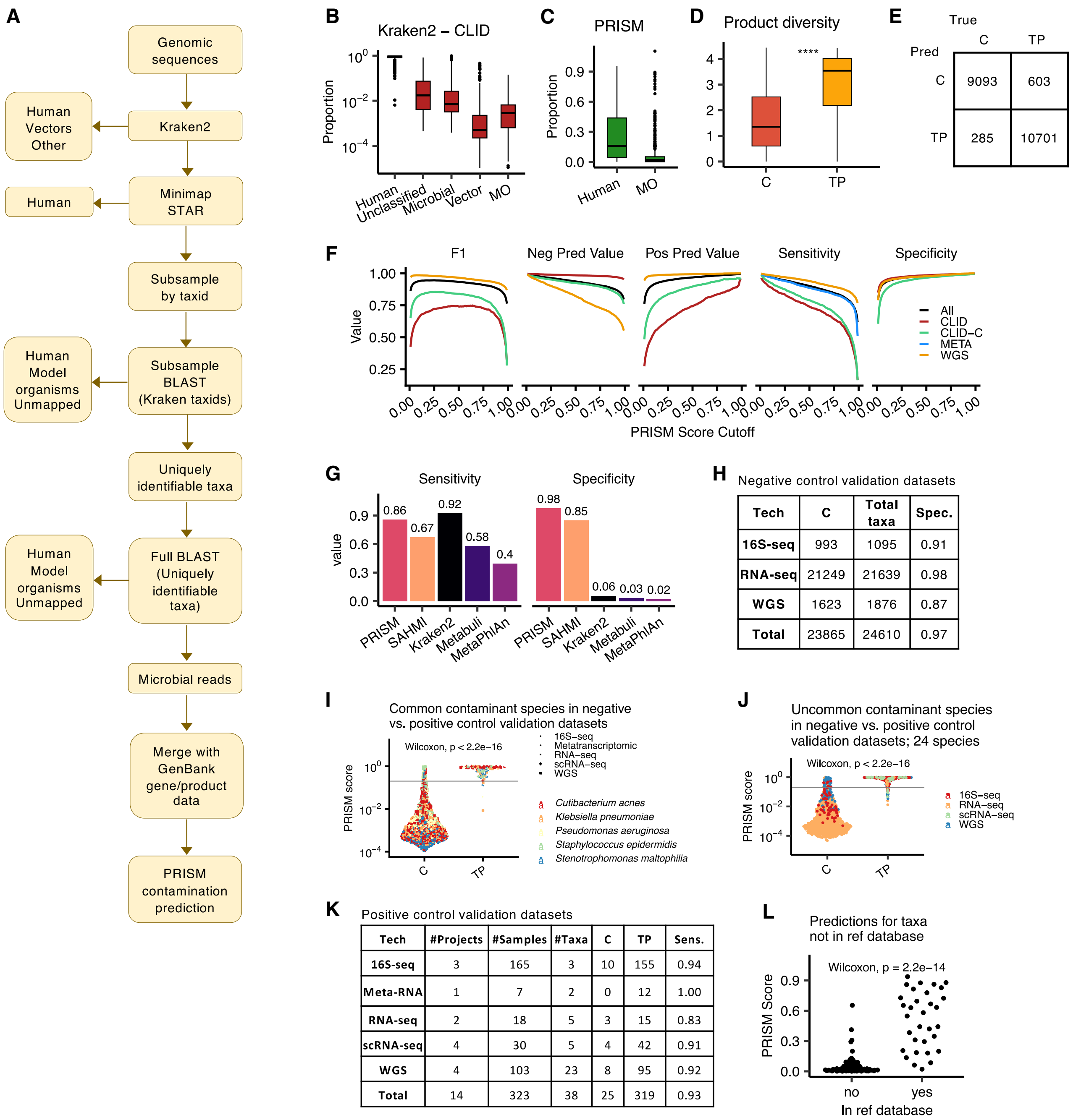
Recent controversy in the cancer microbiome field highlights the need for more reliable microbial detection from human genomic data. Here, we develop PRISM, an efficient computational framework for precise microorganism identification and decontamination from low-biomass sequencing data. PRISM achieves robust performance when benchmarked on 230 independent datasets with known true-positive and contaminant taxa. We then use PRISM to profile 25 cancer types from The Cancer Genome Atlas and Clinical Proteomic Tumor Analysis Consortium. We identify consistent microbial signatures in gastrointestinal tract, head-and-neck, and urogenital tract tumors, and sparse signal elsewhere. In pancreatic cancer, we associate microbial detection with altered host protein glycosylation pathways and greater smoking exposure. Lastly, we consider the impact of sequencing approaches on positive and negative data interpretation. Overall, PRISM improves the reliability of microbial profiling and allows leveraging of existing human genomic data for the concurrent detection of host-microbial signatures with potential molecular and clinical significance.

INTRODUCTION

Despite numerous associations between the microbiome and human health and disease,[1–8] fundamental questions remain about its size, distribution, and presence in various human tissues.[9] This is exemplified by recent controversies around the cancer microbiome, where independent analyses of the same genomic data have reached widely different conclusions.[10–16] For example, many reported microbial sequences in The Cancer Genome Atlas (TCGA) have been attributed to the contamination of microbial databases by human and vector sequences,[16] whereas a reanalysis of TCGA data using a broad range of methodologies continued to identify robust microbial signals.[11] Separately, tissue staining for bacterial lipopolysaccharide (LPS) in breast cancer[17] has not been well-reproduced,[18] and in pancreatic cancer, 16S ribosomal gene sequencing was found to discriminate short and long-term survivors in one study,[19] but was minimally detected in a separate cohort.[20]

Such differences could stem from false negative and false positive results. False negatives arise from variations in source tissues, sample handling, preservation, and processing, and incomplete reference genomes.[21] False positives arise from both contamination and technical artifacts. Contamination originates from whole microorganisms or adventitious microbial biomatter within reagents or from the ambient environment.[22–24] Technical artifacts arise from incorrect reference genomes, wherein contamination by human, vector, or microbial DNA is a known issue,[25,26] as well as from errors in taxonomic classifiers, which often are only tested using simulated datasets due to a lack of real-world benchmarks.[27,28] Gold-standard contamination controls and full-length read alignment against high-confidence microbial references can help mitigate these issues. However, such controls are often missing, and unbiased, full-length alignment of large datasets (e.g., using BLAST[29]) is usually computationally infeasible.

As such, there is a clear need for improved analysis methods to identify true (tissue-resident) microbial signals, especially in the low microbiome biomass setting when profiled by genomic sequencing. Here we develop PRISM (**Pr**ecise Identification of **S**pecies of the **M**icrobiome), a computationally efficient method to reduce technical error and contamination in genomic sequencing data. We benchmarked PRISM using data from 230 separate studies with known microbial burdens and showed that it identifies tissue-resident taxa with high sensitivity and specificity. We then used PRISM to detect microbes in 25 cancer types in TCGA and the Clinical Proteomic Tumor Analysis Consortium (CPTAC). We identified subsets of tumors with rich microbial content, found microbial associations with tumor molecular and clinical characteristics, and explored causes of microbial differences between CPTAC and TCGA. PRISM confirms and substantially improves upon our previous



Pred \ True	C	TP
C	9093	603
TP	285	10701

Negative control validation datasets

Tech	C	Total taxa	Spec.
16S-seq	993	1095	0.91
RNA-seq	21249	21639	0.98
WGS	1623	1876	0.87
Total	23865	24610	0.97

Positive control validation datasets

Tech	#Projects	#Samples	#Taxa	C	TP	Sens.
16S-seq	3	165	3	10	155	0.94
Meta-RNA	1	7	2	0	12	1.00
RNA-seq	2	18	5	3	15	0.83
scRNA-seq	4	30	5	4	42	0.91
WGS	4	103	23	8	95	0.92
Total	14	323	38	25	319	0.93

Figure 1. PRISM testing and validation

(A) Flowchart of PRISM's key steps.

(B) Boxplots of Kraken2 summary results on CLID (n = 515 samples).

(C) PRISM reclassification of Kraken2-identified microbial reads in CLID (n = 515 samples).

(D) Boxplots compare the Shannon diversity of truly present (TP) vs. contaminant (C) microbial products in CLID (n = 20,682 taxa). See Figure S1 for all features and see STAR Methods for specific feature details and formulas. P-value indicates Wilcoxon testing; ****, $p < 0.0001$.

(E) Confusion matrix shows the PRISM model performance on independent test datasets in classifying contaminants (C) and truly present (TP) taxa.

(F) Performance metrics of PRISM across a range of PRISM score cutoffs.

(G) Bar plots compare the sensitivity and specificity of five methods on CLID.

(H) PRISM specificity in predicting contaminants from negative control datasets, using a PRISM score cutoff of 0.5. C, contaminant; Total taxa, total number of taxa detected; Spec., specificity.

(I) Swarm plot compares PRISM scores for five species when they were known contaminants (C) vs. truly present (TP). The gray line marks PRISM score = 0.1. Points are colored by species and shaped by the sequencing type (n = 2104 data points).

(legend continued on next page)

work on SAHMI[30] and pancreatic cancer[31] with increased rigor, direct interrogation of microbial reads, and large-scale, quantitative benchmarking.

RESULTS

Addressing taxonomic misclassification

PRISM provides improved microbial profiling through two key steps: (1) removing technical artifacts (misclassified taxa) and (2) distinguishing truly present species from contaminants (correctly classified but absent from the original sample) using a gradient-boosted tree classifier (Figure 1A).

In developing PRISM, we identified three key sources of taxonomic misclassification specifically encountered using Kraken2,[32] a high performing taxonomic classifier[27,28]: (1) insufficient host read removal (despite including human genome GRCh38 in Kraken2's reference database), (2) multi-mapping reads, whereby many microbial reads have near-identical full-length alignment to multiple taxa, and (3) the presence of sequences from vectors and model organisms that are misclassified as microbial.

To address these, PRISM begins with fast k-*mer*-based taxonomic surveillance using Kraken2[32] and a reference database of complete microbial genomes, human genomes (GRCh38, T2T), common model organisms, and known vector sequences (see STAR Methods for details). It then takes the putative microbial reads and performs additional host-removal using Minimap2[33] and STAR.[34] From the remaining microbial reads, PRISM creates a taxa-representative subsample of the data and aligns it with BLAST[29] against human, model organism, and the remaining microbial genomes. It uses reads that map to only one taxon to identify the species present and then remaps all putative microbial reads to this refined set of species with BLAST and enumerates taxa and gene counts at the highest taxonomic level possible.

To validate that this approach retains known truly present species, we assembled a dataset of 515 human cell-line RNA-seq experiments deliberately infected with microbes from 60 unique species, referred to as the cell-line infection dataset (CLID, Table S1). In each sample, the introduced microbe is truly present, and all other detected taxa are presumed contaminants or artifacts. Kraken2 classified a median of 1% of reads/sample as microbial (Figure 1B), along with a surprisingly high median of 160 taxa/sample, when only 1–4 taxa/sample were expected. Of the Kraken-identified microbial reads, PRISM reclassified 16% (median) as human (most commonly ribosomal or mitochondrial sequences; Table S2) and 2% as model organism (Figure 1C). After accounting for multi-mappable reads (46% of reads from the subsampling step; Figures S1A–S1E), PRISM ultimately identified a median of 14 taxa/sample while retaining all 675 truly present pathogens; a substantial reduction in misclassified taxa while maintaining high sensitivity to the truly present taxa.

Data-driven prediction of truly present vs. contaminant taxa

PRISM uses a gradient-boosted tree model to classify truly present vs. contaminant taxa at the species or strain level while also reporting read counts for higher-level taxa when species-level resolution is not feasible. To train this model, we assembled a large dataset with four main data types: (1) CLID, (2) random *in silico* combinations of CLID (CLID-C), (3) *in silico* combinations of whole genome sequenced bacterial isolates and negative controls (WGS), and (4) metatranscriptomic data (META). In total, our dataset included 833 samples, 416 unique true-positive species, and 1,266 unique contaminants, with a total of 20,892 observed taxa. This training set is notable for its size, variety, and controlled definition of true vs. contaminant taxa (Table S3).

We used these data and their mapping statistics to create 40 features that quantify different aspects of taxon sequence diversity, based on the hypothesis that truly present species yield a greater diversity of mapped reads than contaminants (see STAR Methods, Figures 1D and S1F–S1L). The model predicts a "PRISM score" for each taxon, i.e., the probability that a taxon is a contaminant (0) or is truly present (1). Using 5-fold cross-validation (stratified by project), the model had robust performance across a range of data conditions (overall sensitivity = 0.95, specificity = 0.97, positive prediction value = 0.97, negative predictive value = 0.94; McNemar test $p < 2e-16$; Figures 1E and 1F and S1M-S1Q). We compared PRISM's performance on CLID to four other methods for taxonomic classification or *in silico* decontamination (Kraken2,[32] MetaPhlAn,[35] Metabuli,[36] and SAHMI[30]) and included the additional step of filtering a "blacklist" of 497 common contaminating genera[14] from their outputs. PRISM demonstrated the highest specificity and second-highest sensitivity in detecting truly present microbes, outperforming all other methods overall (Figure 1G), underscoring its reliability and improvement over existing approaches.

Validating PRISM

We next validated the PRISM score using several new datasets and scenarios. First, we tested it on new negative control samples (Table S4) from RNA-seq (uninfected cell lines), WGS, and 16S-rRNAseq experiments (reagent or blank samples) where all taxa are contaminants. PRISM had an overall specificity of 0.97 in identifying contaminants (Figure 1H).

Second, we tested PRISM on new positive control samples (Table S4). Focusing on 5 common contaminants and 24 uncommon contaminants in the negative control data, we analyzed 13 new datasets[37–49] in which these species were validated as truly present via positive cultures from human infections. PRISM scores were significantly higher for these species when they were truly present vs. contaminant across all sequencing types (Wilcoxon $p < 2e-16$; Figures 1I and 1J). PRISM had an overall sensitivity of 0.93 in identifying these validated truly present species (Figure 1K), with excellent performance across a range of data types and relative abundances (Figures S2A–S2D).

(J) Swarm plot similar to (I) but for 24 uncommon species ($n = 824$ data points).

(K) PRISM sensitivity across all of the positive control datasets. C, predicted contaminant; TP, predicted to be truly present; Sens, sensitivity.

(L) Swarm plot of PRISM scores for samples from the CDC-HAI project that contained WGS of taxa that were (yes, $n = 20$) or were not (no, $n = 28$) in our reference database. Boxplots show median (line), 25th and 75th percentiles (box), and 1.5xIQR (whiskers), and individual points represent outliers (B-D). See also Figures S1 and S2 and Tables S1, S2, S3, and S4.

Third, we compared PRISM scores to read counts for their ability to classify truly present species. Logistic regression demonstrated a highly significant effect for PRISM scores (Wilcoxon $p < 2e-16$), whereas read counts alone were not predictive ($p = 0.21$; Figure S2E), highlighting the added value of PRISM scores toward identifying truly present species.

Fourth, we tested whether PRISM scores were biased toward taxa that were more frequent true positives in the training data. We binned taxa into deciles by their number of instances as true-positives in the training data and assessed their PRISM scores in the negative control data. While PRISM scores differed significantly across groups (Kruskal-Wallis, $p < 2e-16$, Figure S2F), median scores were <0.004, indicating minimal classification ambiguity across groups. To test if this variation was due to training taxon frequency, we repeated the analysis using models retrained on balanced data (10, 15, 25 instances per taxon). Using the same decile groupings, we again observed significant score differences (Kruskal-Wallis, $p < 2e-16$, Figure S2G), indicating that training data taxon frequency did not bias PRISM scores on new samples.

Fifth, we tested PRISM when the truly present organism was absent from its reference database. We analyzed WGS data for 48 species from the CDC-HAI project.[50] When the truly present species was in the database, PRISM detected only that species in 29/30 samples (Figure S2H) and classified all as truly present (Figure 1L). When absent, PRISM reported 1–5 species per sample (Wilcoxon $p = 1e-9$, Figure S2H), all from the same genus as the true taxon, often reflecting recent GTDB genus-level monophyletic reclassifications.[51] Reads for these mismapped taxa aligned well (median 94% base match) but showed lower identical base match rates ($p < 2e-16$; Figure S2I) compared to when the truly present species was in the database, demonstrating PRISM's robustness in these scenarios. Collectively, these validation experiments highlight PRISM's strong diagnostic performance across diverse scenarios.

Microbial detection in The Cancer Genome Atlas

We next applied PRISM to detect microbes in WGS data from 25 cancer types in TCGA (n = 2,323 samples, pre-processed with a comprehensive host filtration pipeline[52]; Table S5). PRISM identified abundant microbes in head and neck, gastrointestinal (GI) tract, and cervical cancers, but generally minimal microbes in other cancer types (Figure 2A), where most samples (>90%) had <100 microbial counts per million human reads (CPM; Figure 2B). Even in cervical and stomach cancers, the high total microbial load was driven by a small subset of samples (<10%; Figure 2B).

Examining taxa CPM and PRISM scores revealed that most taxa had extremely low values suggestive of contamination, although a subset likely represented true signals (Figure 2C). Applying conservative thresholds motivated by these distributions to filter likely contaminants (CPM >0.5 and PRISM score >0.1) revealed that outside of oral and GI tumors, most samples had sparse microbial detection (Figure 2D).

In other cancer types, taxa with high PRISM scores and read counts were predominantly from five common contaminant genera: *Escherichia*, *Bradyrhizobium*, *Cutibacterium*, *Saccharomyces*, and *Sphingomonas* (Figure 2E). Beyond these, a few taxa with elevated PRISM scores were detected, suggesting rarer but potentially relevant associations. Taken together, these global observations indicate that in TCGA WGS data, oral and GI tumors harbor reliably detectable microbiomes, whereas in other cancer types, microbial signals are detectable in selected samples, but broader microbiomes are not reliably detectable.

We next used conservative filtering (CPM>0.5, PRISM score>0.1, >1 data center, >10% samples) to highlight the strongest microbe-cancer type associations (Figure 2F; all data in Table S5). Colorectal cancers were dominated by anaerobic gut and oral commensals, including *Bacteroides fragilis*, *Fusobacterium nucleatum animalis (Fna)*, *and Escherichia coli*. Notably, Fna is consistently enriched in colorectal cancer (CRC) and has been linked to tumor progression, metastasis, and poor prognosis. We detected multiple *Prevotella* species in CRC, most prominently *P. multiformis* and *P. intermedia*, the latter of which is particularly enriched in CRC and exhibits additive effects with *Fna* in promoting malignant transformation and metastasis.[53] Additional oral anaerobes such as *Porphyromonas*, *Selenomonas*, and *Parvimonas* were also present, reinforcing the oral-gut microbial overlap characteristic of colorectal cancer.

Species detected in oral, esophageal, and gastric cancers also included gut commensals but were enriched for genera associated with the upper digestive tract and with periodontal disease, such as *Helicobacter*, *Campylobacter*, *Neisseria*, *Capnocytophagia*, *Tannerella*, *Prevotella*, *Porphyromonas*, and *Selenomonas*. These patterns are consistent with prior studies linking gastric and esophageal cancers to oral dysbiosis and colonization of the gastric mucosa.[54,55] In gastric cancer, we also detected Epstein-Barr virus in multiple samples, a known driver and prognostic marker, along with Cytomegalovirus, a known bystander in gastric tissue.

In urogenital tract cancers, we detected several ecologically relevant species. Human papillomavirus (HPV) 16 was prevalent in cervical cancer (and head and neck cancer) and was detected in 5 bladder cancer samples, consistent with its established role in carcinogenesis. Other prominent and ecologically relevant genera included *Peptoniphilus, Anaerococcus, Schaalia* in bladder cancer, as well as 1 case of BK virus, and *Lactobacillus* in uterine and ovarian cancer.

We compared PRISM's results in head and neck, colon, and cervical cancers to a recent reanalysis of TCGA by Ge et al.[16] Without contamination filtering, total microbial read counts per sample were similar between methods (Wilcoxon p = 0.54; Figure S3A). However, only 39% of taxa detected by Ge et al. passed PRISM's multimapping filter (Figure S3B), indicating that PRISM was sensitively identifying microbial reads while more precisely assigning them to taxa (still with near full sequence alignment). As an additional validation, we compared reads assigned to HPV between the two studies and found no significant difference in read counts or relative proportions (Figures S3C and S3D), confirming that PRISM preserved true microbial signal while filtering noise. Collectively, these findings highlight the power of PRISM to identify known and new microbial species in sequencing data.

Microbial detection in the Clinical Proteomic Tumor Analysis Consortium

As an orthogonal validation cohort, we applied PRISM to ribosome-depleted RNA-seq data from CPTAC3 (brain, head

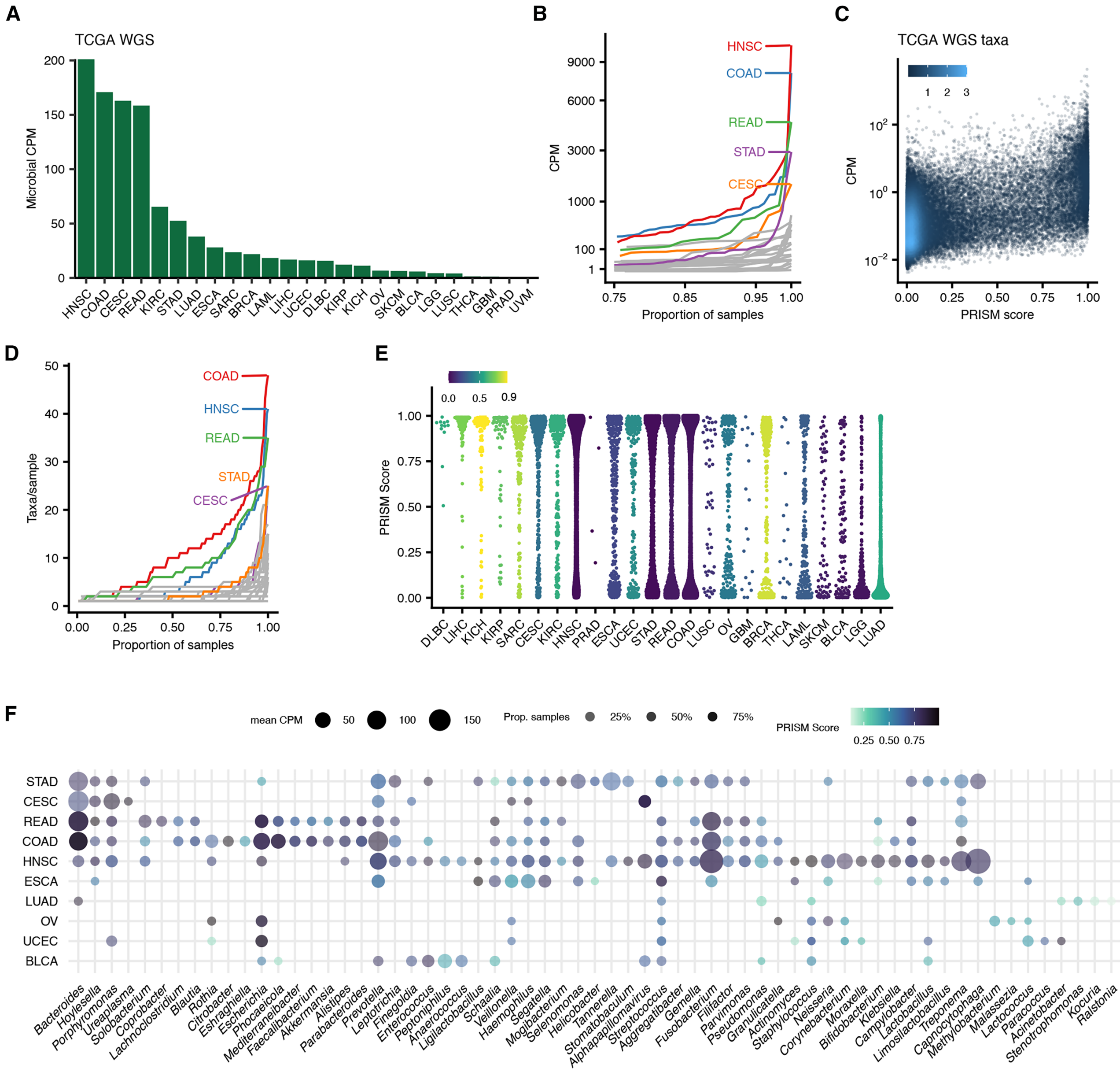


Figure 2. Detection of microbes in TCGA WGS

(A) Total microbial counts per million human reads (CPM) across all tumor samples in 25 cancer types in TCGA as detected by PRISM. See STAR Methods for GDC cancer codes (n = 2323 samples).

(B) TCGA samples ordered by CPM per sample. The x axis is scaled to reflect the proportion of samples in each cancer type.

(C) CPM and PRISM scores for microbial taxa detected in TCGA. Each point is a taxon. The color scale represents relative point density (n = 180423 data points).

(D) Line plot of samples ordered by number of taxa detected per sample for taxa with CPM>0.5 and PRISM score>0.1 (n = 2323 samples).

(E) Swarm plots show the distribution of PRISM scores for taxa in each cancer type for taxa with CPM>0.5. The color bar indicates the proportion of detected taxa that are from these genera: *Escherichia, Bradyrhizobium, Cutibacterium, Saccharomyces,* and *Sphingomonas* (n = total 93238 taxa). The maximum value of the color bar is 0.94.

(F) Dot plot of specific microbial genera detected in TCGA with CPM>0.5, PRISM score>0.1, and >1 sequencing center. See also Figure S3 and Table S5.

and neck, kidney, lung, ovary, pancreas) and poly-A selected RNA-seq data from CPTAC2 (breast, colon, ovary; total 2075 sequencing runs; Table S6). These samples were pre-filtered for alignment to human GRCh38. Despite this, PRISM classified a median of 79% of the remaining reads as human and 2.5% as microbial (Figure 3A). PRISM detected 10°-10^{7} microbial reads/sample, the majority in head and neck tumors (Figure 3B). Most taxa had low PRISM scores, though a subset showed higher scores indicative of likely true microbial presence (Figure 3C). Outside of head and neck and colon cancer, detection across samples was highly sparse (Figure 3D). Minimal filtering for taxa with PRISM score >0.1 and those present in >1 sample

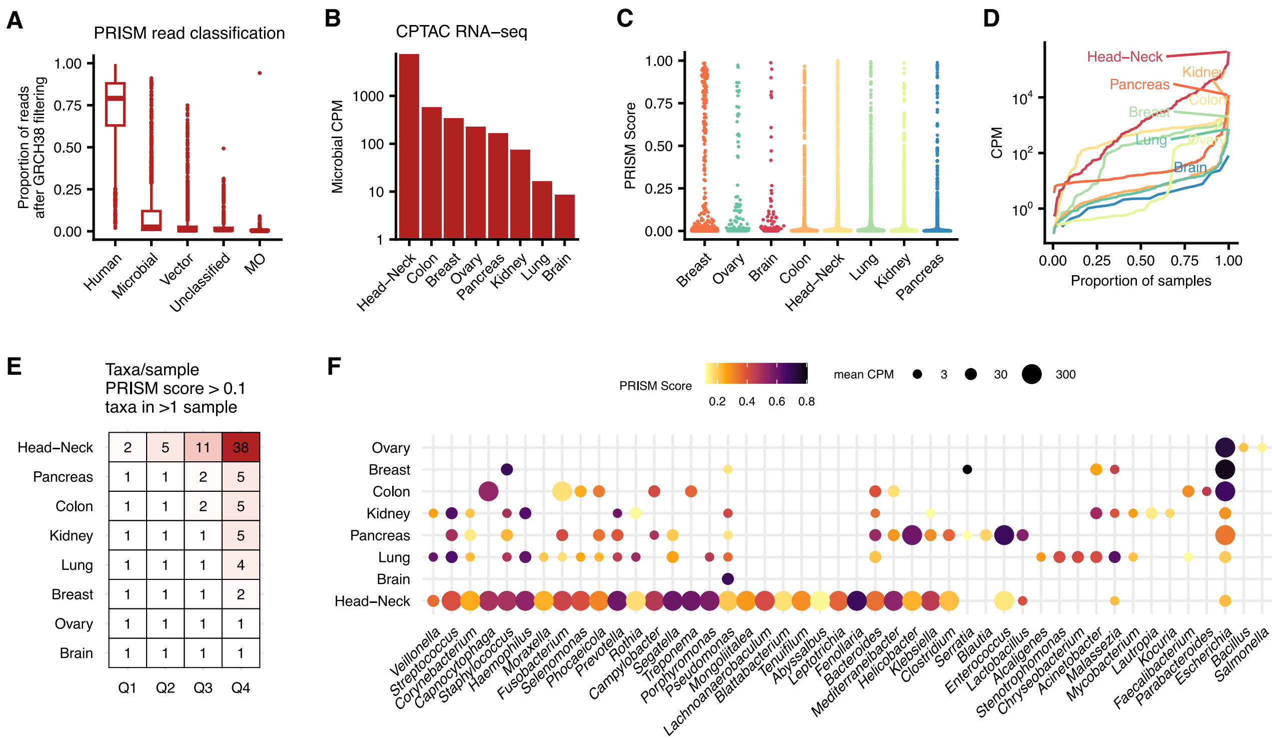


Figure 3. RNA-seq based detection of microbes in CPTAC

(A) PRISM classification results on CPTAC reads that did not align to GRCh38 (n = 2075 samples). Boxplots show median (line), 25th and 75th percentiles (box), and 1.5xIQR (whiskers). Individual points represent outliers.

(B) Total microbial counts per million human reads (CPM) in CPTAC detected by PRISM.

(C) Swarm plots show the distribution of PRISM scores for taxa with CPM>0.5 (n = 13153 taxa).

(D) Line plot of samples ordered by the number of taxa detected per sample for taxa with CPM>0.5 and PRISM score>0.1 (n = 2075 samples).

(E) Table shows the quartile values of the number of taxa per sample for CPTAC samples after taxa filtering for PRISM score >0.1 and taxa in >1 sample.

(F) Dot plot of specific microbial genera detected in CPTAC with CPM>0.5, PRISM score>0.1, and >1 sequencing center. See also Figure S3 and Table S6.

resulted in >75% of all samples (except head and neck) having 2 or fewer taxa (Figure 3E).

Using these minimal filtering criteria to highlight the strongest detectable signals in CPTAC revealed broadly concordant microbial communities compared to TCGA (Figure 3F). Head-and-neck cancers exhibited the richest microbial profiles, dominated by oral anaerobes such as *Veillonella*, *Streptococcus*, *Haemophilus*, *Prevotella*, *Fusobacterium*, and *Porphyromonas*, consistent with prior metagenomic studies.[56,57] Colorectal tumors retained classical gut-oral signatures, including *Fusobacterium*, *Bacteroides*, *Prevotella*, and *Parabacteroides*, reaffirming that these genera are reproducibly detectable even in poly-A-selected libraries. Pancreatic cancer samples included taxa from both the upper and lower gastrointestinal tract, including *Helicobacter* and *Bacteroides*, consistent with prior evidence for microbial translocation in pancreatic adenocarcinoma.[58]

Lung cancer samples harbored several taxa classically associated with oral pathology and chronic airway colonization, such as *Streptococcus*, *Prevotella*, and *Veillonella*.[59] Unexpectedly, we identified the anaerobic gut commensal *Phocaeicola vulgatus* in 9 lung cancer samples (2,325 total reads with PRISM score>0.1). Although this could suggest gut-lung microbial exchange,[60] most *P. vulgatus* reads mapped to a single genomic region corresponding to ribosomal RNA (Figure S3E), raising the possibility of contamination. Notably, *Bacteroides* species were also detected in 5 TCGA lung tumors, but in that case, the reads corresponded to >1,000 unique microbial products, supporting a more plausible biological signal. Other tumor types showed low microbial abundance without consistent ecological enrichment. Overall, the CPTAC data recapitulate the major oral- and gut-derived microbial patterns observed in TCGA WGS despite differences in sequencing approaches.

Technical factors substantially impact microbial detection

Microbial detection varied with technical factors across CPTAC and TCGA datasets. For example, poly-A selection (PAS) in CPTAC2 colon tumors markedly reduced detectable microbial reads compared to ribosome-depleted (RD) head and neck samples in CPTAC3 (Figure 3) and to WGS colon samples in TCGA (Figure 2). Additionally, we attempted to analyze PAS RNA-seq data in TCGA; however, the vast majority of samples had few to no reads, and thus these data were not further analyzed. In CLID, RD also captured more microbial content than PAS (Figures S3F–S3I), confirming its superior sensitivity to microbial reads.

We also assessed the sensitivity of host sequencing to targeted microbial sequencing by comparing PRISM results in

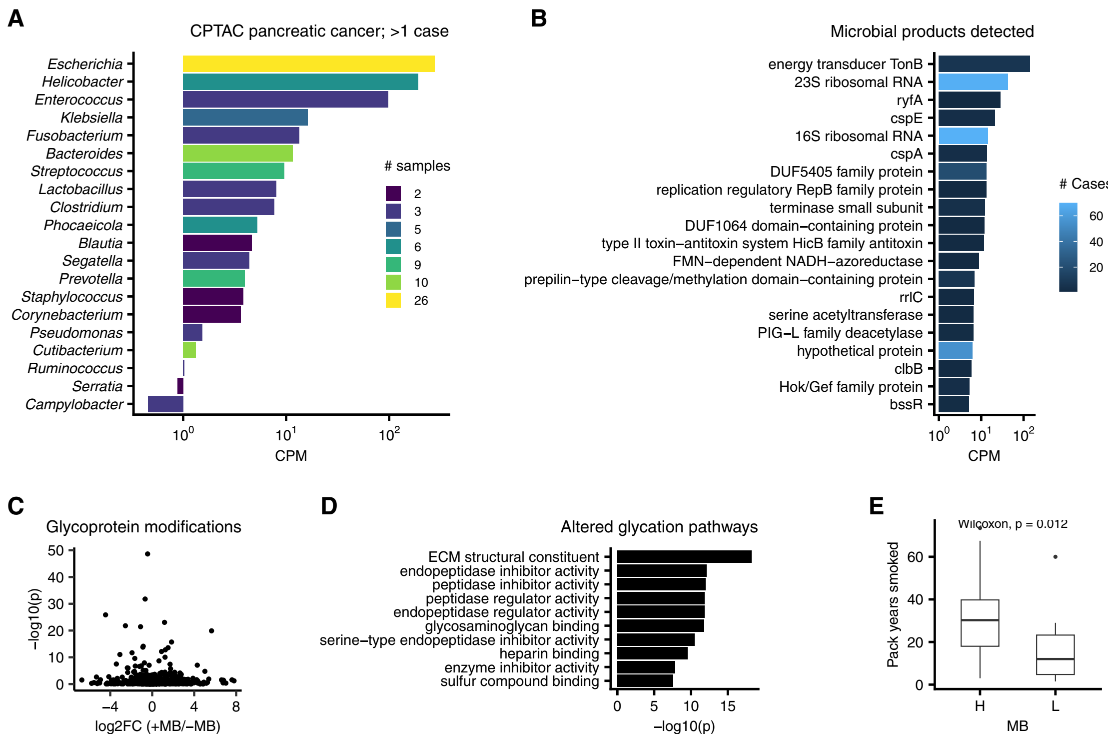


Figure 4. Molecular and clinical associations with the pancreatic cancer microbiome

(A) Bar plot indicates the total counts per million human reads (CPM) for species identified in $n = 155$ cases of pancreatic cancer in CPTAC. Bars are colored by the number of cases in which the species was detected.

(B) Bar plot of the most common microbial products detected in CPTAC pancreatic cancer.

(C) Volcano plot of genes with glycoprotein modification. The x axis is the log2 fold change of average gene modification level in tumors with detectable microbiome (+MB; $n = 62$) vs. those without (-MB; $n = 93$). Y axis indicates the Wilcoxon p-value.

(D) Bar plot indicates gene ontology pathways of the genes with significantly different glycoprotein modifications ($p < 1e-5$) in tumors with vs. without detectable microbiome.

(E) Boxplots of pack years smoked of patients with pancreatic cancer whose tumors have (H, $n = 62$) vs. do not have (L, $n = 93$) detectable microbes using RNA-seq. Boxplots show median (line), 25th and 75th percentiles (box), and 1.5xIQR (whiskers). Points represent outliers. See also Table S7.

four studies[31,61–63] (159 samples) that performed both RNA-seq and 16S rRNA amplicon sequencing on matched samples. Species detected by RNA-seq had higher relative abundances in the 16S data than those detected in 16S but not RNA-seq (Wilcoxon $p < 0.003$; Figure S3J), confirming that RNA-seq preferentially captures high-abundance taxa. These data indicate that the lack of microbial detection in human sequencing data cannot always be presumed to show absence.

Lastly, we tested for batch effects in the TCGA colon and head and neck cohorts using PERMANOVA on Aitchison (CLR) distances, with sequencing or data center as the factor of interest. Only the sequencing center in head and neck tumors showed a modest effect, explaining 6% of the variance ($p = 0.04$); all others were not statistically significant (Figures S3K–S3N). Although batch effects were minimal here, PRISM does not eliminate them, and thus, considering their effect on microbial detection is necessary.

Microbial associations with pancreatic cancer

Lastly, we explored our microbial findings in pancreatic cancer in CPTAC, where there has been recent debate about microbial influence.[19,20,31] The taxa detected by PRISM included oral, intestinal, and opportunistic bacteria previously linked to mucosal inflammation and carcinogenesis (Figure 4A). *Escherichia coli*, the most frequent species, can translocate from the gut to the pancreaticobiliary tract and release colibactin, a genotoxin that induces DNA damage and inflammation.[64,65] *Helicobacter pylori was detected in multiple samples, with prior studies showing mixed evidence for H. pylori infection or seropositivity in pancreatic cancer*.[66,67] Oral taxa such as *Fusobacterium nucleatum*, *Streptococcus anginosus*, and *Prevotella intermedia* are implicated in periodontitis and pancreatic inflammation and may reach the pancreas via hematogenous or GI luminal translocation and promote local inflammation and immune evasion.[68–70] Detected intestinal commensals such as *Bacteroides fragilis* and *Phocaeicola vulgatus* can degrade mucins and alter epithelial integrity.[71]

We examined the microbial genes and products identified by PRISM (Figure 4B; Table S7), with the most abundant being linked to host interaction, stress response, and genotoxicity. The abundant **TonB energy transducer** suggests active bacterial transport and iron acquisition, processes that can support microbial persistence in tumors. The detection of clbB is notable given prior evidence that colibactin-producing *E. coli* induce

DNA damage and promote tumorigenesis.[72] Several stress-response and persistence genes, including HicB and **Hok/Gef**, may facilitate microbial survival under metabolic or immune stress.[72,73] These observations suggest that bacteria within a subset of pancreatic tumors are engaging in survival and possibly tumorigenic activities.

Acknowledging sparsity in the data, we next compared pancreatic tumors with (+MB) and without (-MB) detectable microbes to assess molecular and clinical differences. Microbial presence was not associated with overall mutation burden or global mRNA, protein, phosphorylation, or methylation levels. However, 76 glycoproteins showed significant modification differences between +MB and –MB tumors (adjusted $p < 0.05$; Figure 4C), enriched for pathways involved in extracellular matrix organization, protease inhibition, and glycosaminoglycan binding (Figure 4D). These pathways are particularly relevant in pancreatic cancer, which is characterized by a dense, immune-excluded stroma.[58,74] Lastly, patients with microbial detection had significantly greater smoking history (Wilcoxon $p = 0.01$, Figure 4E), consistent with one recent study.[75] To assess potential confounders, we also evaluated associations with other clinical (e.g., biopsy site, tumor size, alcohol use, BMI) and found no significant associations that explained our findings. While these observations require validation, they converge on biologically plausible mechanisms and highlight the potential for PRISM to identify interesting microbial associations.

DISCUSSION

In developing PRISM, we addressed four major sources of false-positive microbial signals: (1) misclassified human reads, including alternative haplotypes, repetitive mitochondrial and ribosomal sequences, and unplaced scaffolds,[76] mitigated using four host-alignment methods with GRCh38, T2T, and RefSeq references; (2) vector and model organism sequences; (3) multi-mappable microbial reads; and (4) contaminant microbial DNA. We introduced the **PRISM score** and validated its accuracy in curated datasets. Although not error-free, at least, PRISM substantially reduces technical artifacts and enables robust reanalysis of genomic data.

In TCGA and CPTAC, we identified coherent microbial signatures in head and neck and gastrointestinal tract tumors, detectable microbes in subsets of urogenital tract tumors, and a sparse microbial signal in other cancer types in these datasets. Our findings are consistent with recent studies reassessing microbial content in pan-cancer datasets,[11,16,77] which report high microbial content in the same tumor types and varying levels in others. However, we caution against over-interpreting negative results in the absence of targeted microbial assays. For example, we found minimal microbes in lung cancer in TCGA and CPTAC, whereas metagenomic approaches have repeatedly demonstrated lung intra-tumoral bacteria[17] associated with smoking and mutation status.[78]

In the CPTAC pancreatic cancer cohort, microbes were associated with altered protein glycosylation, consistent with evidence linking glycosylation changes to altered host-microbiome interactions and immune dysregulation in inflammatory bowel disease and some cancers.[79,80] Whether these microbes alter the tumor glycome or an altered glycome attracts microbes is an important open question. Collectively, these findings suggest that despite varying abundances of tumor-associated microbes, their presence may carry functional and clinical significance in select cancers, warranting further examination.

The key utility of PRISM lies in its capacity to leverage existing human genomic data for concurrent and efficient detection of rare microbial signals, utilizing full-length, gene-level mapping. This enables joint host-microbiome analyses to identify hypothesis-generating molecular and clinical correlates initially without costly, additional laboratory processing. While not replacing gold-standard methods, PRISM opens the potential for reanalyzing substantial troves of existing tumor genomic data to gain insights into host-microbiome interactions.

Limitations of the study

Our study has several strengths and limitations. PRISM was developed and validated on a distinctively large, curated dataset, providing confidence in its robustness. It directly and efficiently interrogates reads, addressing root issues of mapping uncertainties, and it is adaptable to diverse sequencing types and source samples. However, it also has limitations. First is the absence of a universal PRISM score cutoff, as interpretation depends on context, such as taxa counts and score distributions. In this article, we used 0.1 to highlight data in the figures, not as a universal cutoff. It is possible for filtering to remove true microbes with low PRISM scores. This occurred in the validation experiments, where overall sensitivity was 0.93. Second, as PRISM reports uniquely identifiable species, it could miss highly similar species whose reads are multi-mappable with high fidelity. Third, results depend on the inclusion, accuracy, and sampling of reference genomes. Fourth, the PRISM score relies on microbial capture patterns observed in laboratory settings; these patterns may differ in complex microbial communities in human tissues. Fifth, PRISM removes reads that map to human sequences using BLAST, some of which are labeled as "chimeric clones"; this conservative approach risks removing reads of true microbial origin. Sixth, the PRISM score favors a higher diversity of reads; it may be artificially low for species that are dormant or are less characterized in reference databases. Seventh, PRISM is not designed to detect uncharacterized taxa; however, it may highlight reads with unusual mapping patterns that could warrant closer investigation. Eighth, the analysis done on pancreatic cancer in this article was limited to one cohort.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Bassel Ghaddar.

Materials availability

This study did not generate any new reagents.

Data and code availability

All datasets analyzed in this study are listed in Tables S3 and S4. Full accession numbers and references are available in Table S9. All datasets generated in this study are available on the PRISM GitHub: https://github.com/sjdlabgroup/PRISM and from the authors upon reasonable request. The PRISM pipeline is available on GitHub: https://github.com/sjdlabgroup/PRISM and at Zenodo: https://zenodo.org/records/17613853 with https://doi.org/10.5281/zenodo.17613853.

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AUTHOR CONTRIBUTIONS

B.G. conceived the study and designed and performed all analyses. B.G., M.J.B., and S.D. interpreted the data and wrote and revised the article.

DECLARATION OF INTERESTS

BG and SD have jointly filed U.S. Provisional Application No. 63/705,994 on the basis of this work.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT for article editing. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- METHOD DETAILS
  - PRISM algorithm
  - Assembling the PRISM datasets, training the contamination model, and benchmarking
  - CPTAC and TCGA analyses
- QUANTIFICATION AND STATISTICAL ANALYSES

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
TCGA WGS, human filtered reads	GDC, Rob Knight Lab, UCSD	https://api.gdc.cancer.gov/
CPTAC RNA-seq	GDC	https://api.gdc.cancer.gov/
Public genomic datasets	Public Genomic databases	See Table S9 for dataset accessions and references
Software and algorithms		
PRISM	This paper	https://github.com/sjdlabgroup/PRISM https://doi.org/10.5281/zenodo.17613853
R (v4.4.2)	R CRAN	https://www.r-project.org/
Kraken2	Github	https://github.com/DerrickWood/kraken2
STAR (v2.7.11)	Github	https://github.com/alexdobin/STAR
Minimap2	Github	https://github.com/lh3/minimap2
BLAST	Github	https://github.com/ncbi/blast_plus_docs

METHOD DETAILS

PRISM algorithm

Overview

PRISM is a modular computational framework for identifying truly present microbial species from genomic or transcriptomic sequencing data. It integrates multiple alignment and filtering strategies with probabilistic modeling to distinguish truly-present microbial signals from false positives arising from misclassification artifacts and contamination. We describe the reference databases used in this manuscript below, but users may choose to use other custom or strain-enriched databases. The key steps of the pipeline are detailed below.

(1) Initial microbial surveillance with Kraken2. Raw sequencing reads are first surveyed with Kraken2 using a comprehensive reference database containing complete microbial genomes (bacteria, fungi, and viruses), the human reference genomes, known vector sequences, and common model organisms. The human genomes include both GRCh38 (chromosomes 1–22, X, Y, MT; NCBI accession: PRJNA31257) and T2T-CHM13v2.0[81] (NCBI accession: PRJNA559484), the first truly complete, telomere-to-telomere human genome assembly that provides full coverage of previously unresolved genomic regions. The model organisms included were *Mus musculus* (taxid: 10090), *Drosophila melanogaster* (taxid: 7227), *Caenorhabditis elegans* (taxid: 6239), *Danio rerio* (taxid: 7955), *Arabidopsis thaliana* (taxid: 3702), *Rattus norvegicus* (taxid: 10116). This rapid k-mer–based classification step provides an initial microbial profile and removes the majority of host and other non-microbial reads. Kraken2 is chosen for this step for its sensitivity, speed, and ease of use.

(2) Additional host-read depletion with Minimap and STAR. To further reduce host-derived false positives, PRISM employs additional alignment steps. First, microbial reads extracted from Kraken2 are aligned against the human T2T genome using Minimap2.[33] Second, STAR[34] is used to eliminate any additional host-mapping reads from transcriptomic data, where spliced or repetitive elements may evade initial depletion. Human-mappable reads are removed from the putative microbial data.

(3) Taxon-representative subsampling and BLAST alignment. Following host-read removal, PRISM constructs a putatively taxonomically-representative subsample of microbial reads to accelerate subsequent alignment and taxon evaluation. The subsampled reads are aligned with BLAST against the "core_nt" reference database comprising the human genome (NCBI taxid: 9606), common model organism genomes (the same as those included in the Kraken2 step), and microbial genomes identified by Kraken2. This competitive mapping identifies (i) remaining missed human or model organism reads, (ii) reads with no or poor full length mapping, and (iii) reads that map to one or microbial taxa. To increase computational efficiency, PRISM has an option to construct a temporary custom BLAST database restricted to the taxa of interest.

(4) Full-dataset BLAST to uniquely identifiable taxa and GenBank annotation. The reads from the subsampled data that map to only one species are used to determine which microbial taxa are uniquely identifiable. The entire set of microbial reads is then realigned with BLAST to this refined set of microbial taxa, along with human and model organism genomes. PRISM filters read alignments based on alignment quality, percent identity, and coverage to remove spurious mappings. Reads mapping to the human genome or model organisms are excluded, and those mapping to microbial species are retained. For each read, PRISM assigns the most specific possible taxonomic label, i.e., those that map to only one taxon have a species annotation,

and those that map to >1 taxon will have a higher-level annotation. PRISM then maps the aligned microbial reads to annotated genes and products in the GenBank database, producing read-level annotations linking microbial identity with functional genomic context.

(5) Machine learning–based contamination prediction. PRISM computes a structured set of 40 features describing each detected species or strain level taxon. These features are input into a pretrained XGBoost model that outputs a PRISM score, i.e., a probability reflecting the likelihood that each species is truly present rather than a contaminant. These features fit broadly into 5 classes.

The first class of features quantifies taxon gene-product diversity using the BLAST results and is based on the observation that truly-present species express a higher diversity of genes compared to contaminants. These include "fprod", the proportion of BLAST products in the sample assigned to the taxon; "fuprod", the proportion of unique products in the sample assigned to the taxon; "fugene", the taxon's proportion of unique genes in the sample; "prod_div", the Shannon diversity of the taxon's product counts; "gene_div", the Shannon diversity of the taxon's gene counts. Shannon diversity is calculated as $SD = -\sum p_i \ln(p_i)$, where p_i is the proportion of genes or products attributed to gene or product *i.* Because the Shannon diversity correlates with the total number of observations, PRISM does 10 iterations of randomly sampling 100 genes or products and calculating the Shannon diversity, then averages the 10 diversity measures.

The second class of features relates to read multi-mappability and is based on the observation that truly-present species have more uniquely mapping sequences than contaminants. These features are: "ratio", the proportion of taxon's reads that are multimappable; "sacc_ratio4", the ratio of the number of accessions only mapped by uniquely mapping reads to the number of accessions mapped by both uniquely and multi-mapping reads; "sacc_ratio5", the ratio of the number of accessions only mapped by multimapping mapping reads to the number of accession mapped by both uniquely and multi-mapping reads.

The third class of features is based on Kraken2's read count estimates and is based on the observation that truly-present species have more total and unique reads than contaminants, as estimated by Kraken2. The features are: "n2", the total reads assigned to the clade rooted at the taxon; "n3", number of reads specifically assigned to the taxon; "uniq", estimated number of unique k-mers assigned to the taxon; "fmicro", taxon read counts relative to the total reads counts assigned by Kraken to bacteria, fungi, and viruses. While some of these feature class reflect read abundance, which is a technical factor, we note that many PRISM features are derived from normalized proportions or ratios rather than absolute counts, which reduces sensitivity to compositional effects.

The fourth class of features is "k-mer taxonomy", or the proportion of k-mers per taxon assigned to each taxonomic level by Kraken2. These features are based on the observation that truly-present species have more reads assigned to the species level than contaminants. Levels for which k-mer taxonomy proportions are computed include kingdom, phylum, class, order, family, genus, species, unclassified, and host (human).

The fifth class of features relates to misclassifications by Kraken2 and is based on the observation that Kraken2 assigns more reads to taxa related to truly-present species than contaminants. For the kingdom (k_), phylum (p_), class (c_), order (o_), family (f_), genus (g_), and species (s_) levels, PRISM determines the proportion of reads (_r), unique k-mers (_u), and species (_n) in the sample that are assigned by Kraken2 to the taxonomic lineage of each taxon.

(6) Output, computational design, and performance. PRISM summarizes results across multiple resolutions. It reports (i) per-species read counts and PRISM scores, (ii) read-level alignments annotated with gene and product information, (iii) counts of other taxonomic ranks and the least common ancestor of each read, (iv) fasta files containing validated microbial reads, and (v) all output from Kraken2. The PRISM workflow is designed for both accuracy and efficiency. By performing rapid k-mer–based screening and multiple host-removal steps, followed by subsampling and targeted BLAST searches, PRISM substantially reduces computational load while preserving sensitivity. PRISM run time scales with the number of reads and taxa, with an average speed of ∼33,000 microbial reads per hour (Figure S3O). The combination of multiple host-removal stages, competitive alignment, and probabilistic scoring minimizes false-positive microbial calls, enabling reliable detection of truly present microbial species even in host-dominated sequencing datasets.

Assembling the PRISM datasets, training the contamination model, and benchmarking

The cell line infection dataset (CLID) was used for the initial development of PRISM. These were RNA-seq experiments of human cell-lines that had been purposefully infected with a known pathogen. These datasets were identified by querying the Sequence Read Archive for "(((("public"[Access]) AND "rna seq"[Strategy]) AND "transcriptomic"[Source]) AND cell line) AND "Homo sapiens"[orgn:txid9606]" and searching for specific microbial species names. All samples were manually verified that they met our desired criteria. In each CLID sample, the known pathogen(s) were truly-present (each sample had 1-4 known pathogens), and all other taxa detected were contaminants. These datasets are listed in Tables S1 and S3.

To train PRISM's contamination model, we combined CLID with 4 other datasets that we generated which had a larger number of known truly-present species. The first dataset consisted of synthetic combinations of CLID which we called CLID-C. For each CLID-C sample, between 100 and 1000 known pathogen reads from different CLID samples were randomly sampled and combined with reads from known CLID contaminants. In total, we created 153 new CLID-C samples; each sample had between 4 and 45 truly-present species and between 1 and 201 contaminants, all from RNA-seq data. To prevent data leak during model cross-validation, we created CLID-C samples in 5-folds, such that there was no crossover of data from SRA projects amongst folds, i.e., all data from the same SRA project were in the same fold. These samples are denoted by projects m1-m5 in Table S3. Details of their sample origins and taxa are in Table S8.

We also created a dataset called WGS which contained *in silico* combinations of cultured and whole genome sequenced bacterial isolates from human gut,[36] bladder,[37] and the FDA ARGOS database.[38] For these data, from each original sample we randomly sampled reads assigned to the cultured species by Kraken2. We then randomly combined these into 236 new WGS-C samples, each with 1–104 truly-present species. Samples were created in 5-folds, with 2-folds each for the gut and bladder data and 1-fold for the FDA ARGOS data, with no crossover between samples across the folds. These samples are denoted by projects f1-f5 in Table S3. Details of their sample origins and taxa are in Table S8. The WGS dataset also included negative control WGS data for $n = 79$ samples; these were reagent or blank controls from whole metagenome sequencing studies.[39–44]

Lastly, we added data from metatranscriptomic studies from human gut,[45] mouth,[46] skin,[55] and vagina,[47] which we called META. To avoid including contaminants from these datasets, we used Kraken2 to identify and extract reads from taxa with reads greater than the median reads/taxa in its sample. These samples were grouped into 4-folds, one for each study. References for all training and testing datasets are included in Table S9.

We explored these data extensively and created the features discussed in the PRISM algorithm section. These were created heuristically as we searched for metrics that distinguished truly-present and contaminant taxa. We computed those features for our dataset and then used xgboost[82] to train a gradient-boosted decision tree model to classify truly-present vs. contaminant taxa. The model was evaluated using 5-fold cross-validation and the predesigned folds discussed above. The model used the following parameters: nrounds = 40, eval_metric = 'auc', objective = 'binary:logistic'. Although these parameters were chosen, they did not have a substantial impact on the final results over other parameter choices (data not shown).

We benchmarked PRISM's cross-validation performance on the CLID dataset against three other high sensitivity or specificity methods for taxonomic classification (Kraken2,[32] MetaPhlAn,[35] Metabuli[36]) and one method for computational decontamination (SAHMI[30]). All methods were run with default parameters (for SAHMI, we used a negative control percentile cutoff of 0.95). We included an additional step of filtering a "blacklist" of 497 common contaminating genera[14] from the output of these methods (but not from PRISM). Doing this did not reduce sensitivity of any method but slightly increased their specificity.

CPTAC and TCGA analyses

All CPTAC fastq files and clinical metadata were obtained directly from the Genomic Data Commons,[83] and sequencing reads unmapped to the human reference were extracted from the BAM files for analysis with PRISM. From CPTAC, we downloaded RNA-seq data of tumor tissues from brain, breast, colon, head & neck, lung, ovary, and pancreas cancers (CPTAC3 and CPTAC2 datasets; endometrial cancer was not analyzed). Molecular (RNA-seq, mutation, proteomic, phosphoproteomic, glycoproteomic) data from CPTAC pancreatic cancer were obtained from the relevant publication.[84] All data were processed using the standard PRISM pipeline. All raw taxa count data are reported in Table S6. For the pancreatic cancer microbiome association testing with molecular data, we did Wilcoxon testing with individual gene mRNA, protein, phosphoprotein, and glycoprotein levels and with total number of tumor mutations. The gene ontology analysis for the significantly altered glycoproteins was done using the clusterProfiler[85] R package with default parameters; the full human proteome was used as the gene background for this analysis.

Whole genome sequencing TCGA data was obtained from the Rob Knight lab. These samples were preprocessed using a comprehensive host filtration pipeline.[52] Briefly, data were host-filtered using Minimap2 against GRCh38, T2T-CHM13v2.0, and the 94 available pangenomes from HPRC[86] using a pangenome index that includes GRCh38, T2T-CHM13v2.0, all 94 HPRC genomes, and GENCODE v47 and index-based filtration. The resulting host-filtered reads were then analyzed with PRISM with default parameters.

To assess potential batch effects on microbial composition, we calculated Aitchison distances on centered log-ratio (CLR)–transformed microbial count data (pseudocount = 1 to avoid log(0)). Euclidean distances in CLR space (Aitchison distance) were computed and used to perform permutational multivariate analysis of variance (PERMANOVA; 9,999 permutations) with sequencing or data-submitting center as the factor of interest. Principal coordinates analysis (PCoA) based on these Aitchison distances was then used to visualize sample clustering and potential batch-driven structure. Data from all samples and taxa were included.

For the experiment comparing relative abundance in 16S rRNA amplicon data in species that were detected vs. were not in host RNA-seq of the same tissues, we analyzed data from four studies (pancreatic cancer,[31] colorectal cancer,[61] gastric cancer,[63] and colonic tissue without detectable malignancy[62]). Species assignments and counts were obtained using PRISM with default parameters.

QUANTIFICATION AND STATISTICAL ANALYSES

All statistical analyses were performed using R version 4.3.1. All *p*-values were corrected for false-discovery rate (fdr) for multiple hypotheses using the p.adjust function with method = "fdr", unless otherwise stated. The tidyverse package was substantially used for data processing (https://www.tidyverse.org/). The ggpubr package (https://github.com/kassambara/ggpubr) was used to compare group means with nonparametric tests and to perform multiple hypothesis correction for statistics that are noted in the figures. P-values reported as $<2 \times 10^{-16}$ result from reaching the calculation limit for the native R statistical test functions and indicate values below this number, not a range of values.

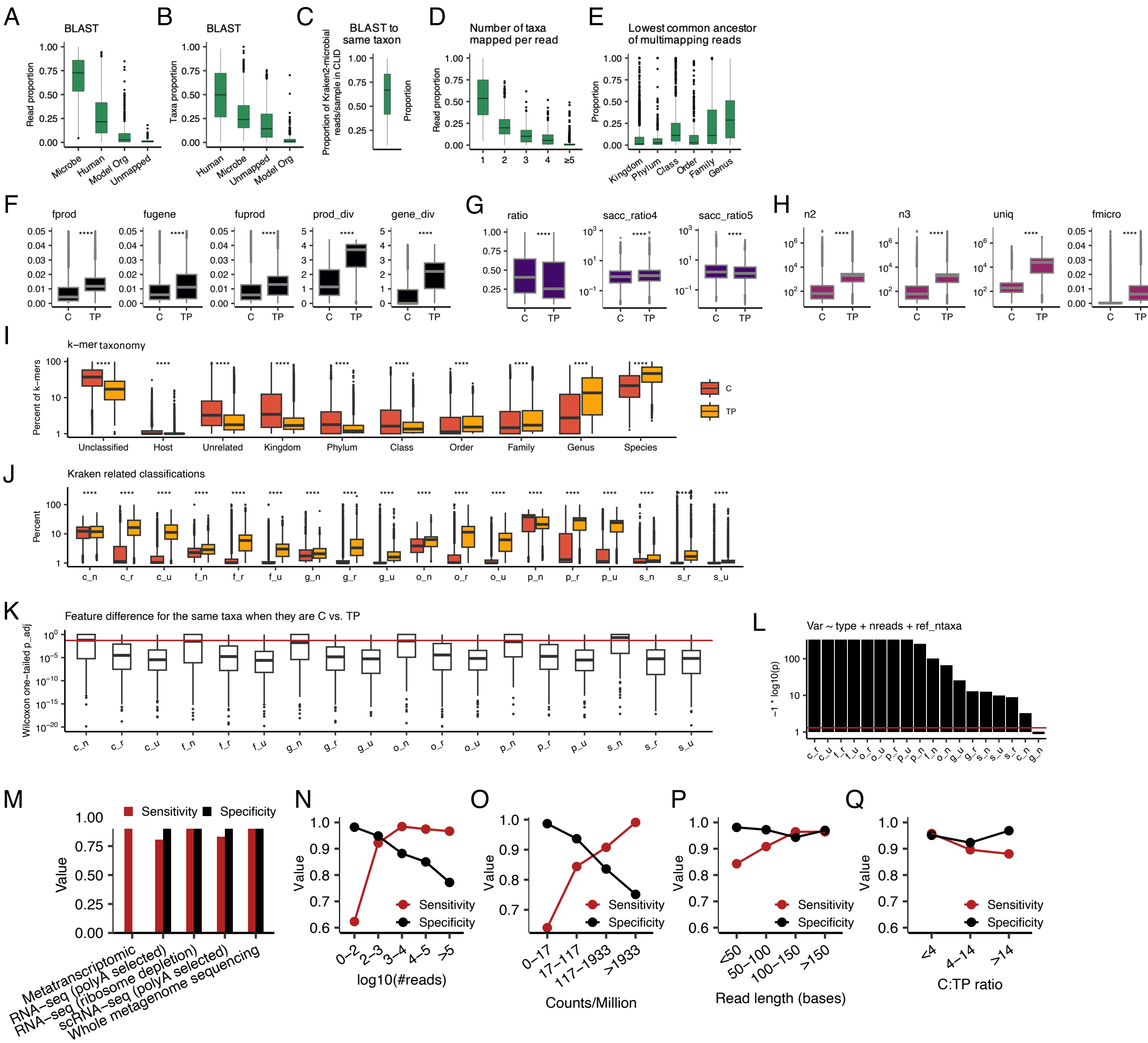
Cancer Cell, Volume *44*

Supplemental information

Reliable detection of Host-Microbe Signatures in cancer using PRISM

Bassel Ghaddar, Martin J. Blaser, and Subhajyoti De

Figure S1

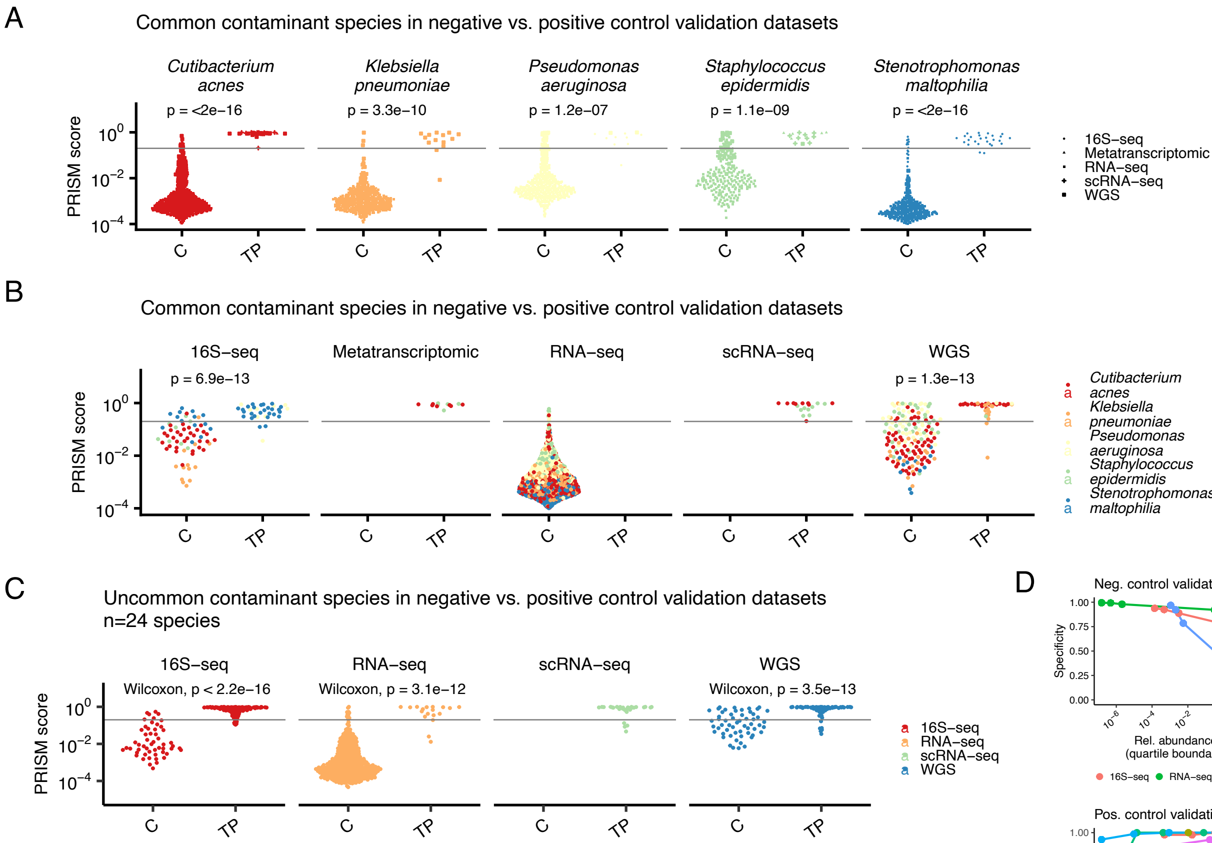


A
BLAST
Read proportion
Microbe
Human
Model Org
Unmapped
B
BLAST
Taxa proportion
Human
Microbe
Unmapped
Model Org
C
BLAST to same taxon
Proportion of Kraken2-microbial reads/sample in CLID
Proportion
D
Number of taxa mapped per read
Read proportion
E
Lowest common ancestor of multimapping reads
Proportion
Kingdom
Phylum
Class
Order
Family
Genus
F
fprod
fugene
fuprod
prod_div
gene_div
G
ratio
sacc_ratio4
sacc_ratio5
H
n2
n3
uniq
fmicro
I
k–mer taxonomy
Percent of k–mers
Unclassified
Host
Unrelated
Kingdom
Phylum
Class
Order
Family
Genus
Species
C
TP
J
Kraken related classifications
Percent
c_n c_r c_u f_n f_r f_u g_n g_r g_u o_n o_r o_u p_n p_r p_u s_n s_r s_u
K
Feature difference for the same taxa when they are C vs. TP
Wilcoxon one–tailed p_adj
L
Var ~ type + nreads + ref_ntaxa
−1 * log10(p)
M
Sensitivity
Specificity
Value
Metatranscriptomic
RNA–seq (polyA selected)
RNA–seq (ribosome depletion)
scRNA–seq (polyA selected)
Whole metagenome sequencing
N
log10(#reads)
O
Counts/Million
P
Read length (bases)
Q
C:TP ratio

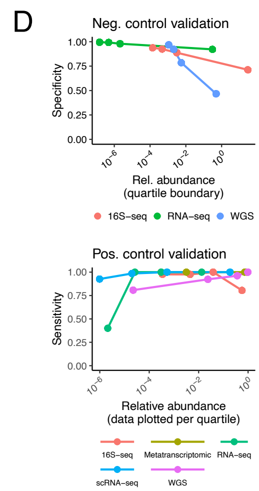
Figure S1. PRISM method benchmarking, Related to Figure 1. (A) Boxplots indicating the proportion of reads and taxa **(B)** per sample in CLID that mapped to each domain using BLAST. These are reads and taxa that were classified as microbial by Kraken2 and then not human by Minimap2 and STAR (n=515 samples). **(C)** Boxplot indicating the proportion of Kraken2-identified microbial reads per sample in CLID that were mapped to the same taxon with BLAST. **(D)** Boxplots indicating the proportion of reads per sample in CLID that mapped to 1 or more taxa with BLAST. X-axis indicates the number of taxa mapped by a given read **(E)** Boxplots indicating proportion of reads per sample in CLID that map to multiple species with BLAST that are resolvable to each lowest common ancestor. Panels A-E indicate that although Kraken2 consistently ranks among the best-performing taxonomic classifiers, a large proportion of taxa it reports in low biomass settings are false positives driven by human and microbial multimapping artifacts. In particular, host filtration with GRCh38 and Kraken2 alone is insufficient. (**F-J**) Boxplots comparing the values of PRISM model features between known contaminant (C) taxa and truly-present (TP) taxa in the CLID datasets. See Methods for specific feature details and formulas. **(F)** Features relatedto microbial and gene and product (as reported by BLAST) statistics calculated from BLAST output; fprod, proportion of total products; fuprod, proportion of unique products; fugene, proportion of unique genes; prod_div, products Shannon diversity; gene_div, genes Shannon diversity. **(G)** Statistics per taxon related to the proportion of reads that are uniquely vs. non-uniquely mappable. mb_strand, strandedness (right vs. left) of mapped reads; mb_uniq, number of unique BLAST accession IDs mapped. **(H)** Statistics per taxon calculated from the Kraken2 report. n2, total reads assigned to the clade rooted at the taxon; n3, number of reads specifically assigned to the taxon; uniq, estimated number of unique k-mers; fmicro, taxon read counts relative to the total reads counts assigned by Kraken to bacteria, fungi, and viruses. **(I)** Proportion of k mers per taxon that Kraken2 assigned to each taxonomic class. **(J)** For each taxon, the number of reads (_n), unique k-mers (_u), or species (_n) that Kraken2 assigned to the same taxonomic lineage for each level of the taxonomic tree. (**K**) Boxplots show one-tailed Wilcoxon p-values testing for a difference in Kraken2-related classification feature values for the same taxa when they were contaminants vs. truly-present. P-values were adjusted for false-discovery. The red line marks p=0.05. (**L**) We used multiple linear regression models to test for the significance of the Kraken-related classification features while controlling for taxon read abundance and the number of related taxa (belonging to the same class) present in the reference database. Bars indicate the coefficient p-value for the tested feature. The model tested feature ~ taxon type (contaminant vs. truly present) + read count + number of same class taxa in the reference database. The red line indicates p=0.05. **(M-Q)** PRISM performance across a range of metrics. **(M)** Bar-plot indicating the PRISM model performance stratified by sequencing type. **(N)** Sensitivity and specificity of the model for taxa with varying numbers of reads. **(O)** Sensitivity and specificity across a range of taxon relative abundance (counts per million total reads). (**P**) Sensitivity and specificity across different sequencing read lengths. **(Q)** Sensitivity and specificity across samples with varying ratios of contaminants to truly-present taxa (C:TP). In all panels, boxplots show median (line), 25th and 75th percentiles (box) and 1.5xIQR (whiskers), and points represent outliers. p-values represent Wilcoxon testing; ****, p<0.0001.

Figure S2

A



D



E

Logistic regression model using datasets from Fig. 2c-d:
Type [contaminant vs. truly-present] ~ PRISM score + read count

Coefficients:

	Estimate	Std. Error	z value	Pr(>lzl)
(Intercept)	-4.86E+00	2.38E-01	-20.425	<2e-16
read count	2.36E-05	1.88E-05	1.252	0.211
PRISM score	8.16E+00	5.28E-01	15.441	<2e-16

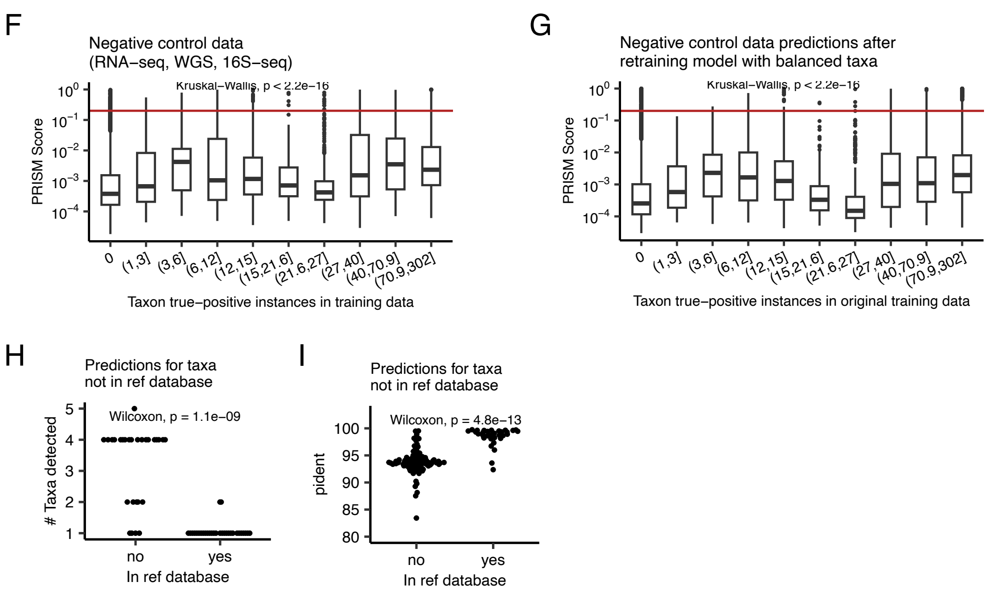
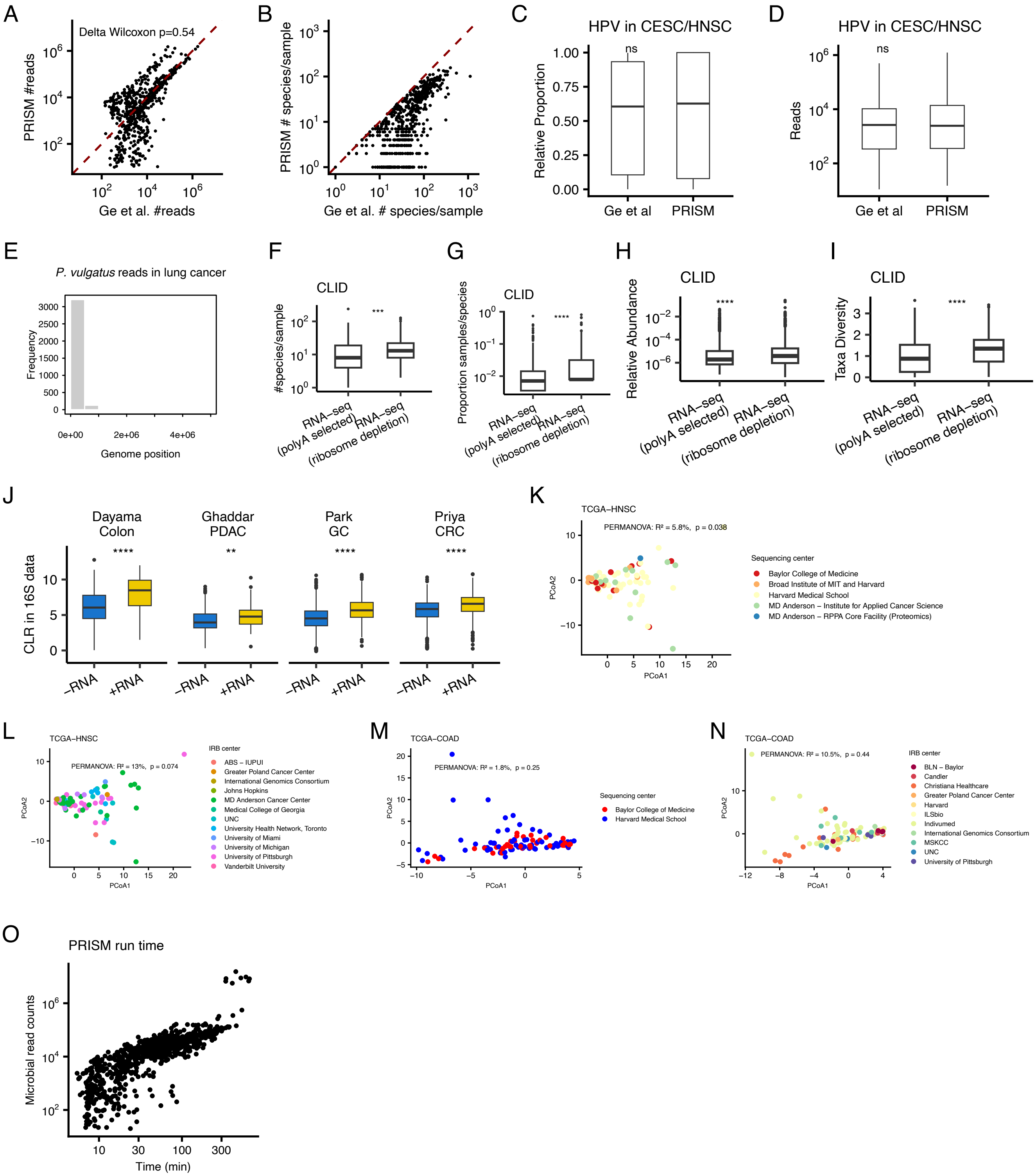


Figure S2. Validating the PRISM score, Related to Figure 1. (**A**) Swarm plots of PRISM scores for the data from Fig. 2K, stratified by species; p-values indicated Wilcoxon testing. (**B**) Swarm plots of PRISM scores for the data from Fig. 2p, stratified by sequencing technology; p-values indicated Wilcoxon testing. (**C**) Swarm plots of PRISM scores for the data from Fig. 2o, stratified by sequencing type; p-values indicated Wilcoxon testing. (**D**) Sensitivity and specificity of PRISM predictions related to the negative control (top) and positive control validation datasets as a function of taxon relative abundance (reads/total reads), stratified by sequencing type. (**E**) Logistic regression modeling of the data from Fig. 2p assessing the significance of the PRISM score vs. read count in distinguishing contaminant vs. truly-present species. **(F)** Boxplots of PRISM scores for the RNA-seq data from cell lines in the absence of known infection, a negative control validation dataset to test for model prediction bias toward over-represented true-positive taxa in its training data. Data are stratified into deciles related to the number of instances in the training data for each true-positive taxon. The red line marks 0.1, the filtering threshold used for plotting in this paper. Boxplots show median (line), 25th and 75th percentiles (box) and 1.5xIQR (whiskers), and points represent outliers. (**G**) Repeating the analysis from (F) but after subsampling the training data to include 15 instances of each contaminant or each truly-present taxon. The new model is then used to predict PRISM scores for the negative control dataset. Taxa are stratified into the same deciles as in (F). Boxplots are as in (F). (**H**) Swarm plot of the number of taxa detected in CDC-HAI WGS of human pathogens for taxa that were (yes, n=20) or were not (no, n=28) in our reference database. (**I**) Swarm plots of percent identical base match (pident) from BLAST for taxa that were or were not in the reference database.

Figure S3



A
Delta Wilcoxon p=0.54
PRISM #reads
Ge et al. #reads
B
PRISM # species/sample
Ge et al. # species/sample
C
HPV in CESC/HNSC
ns
Relative Proportion
Ge et al
PRISM
D
HPV in CESC/HNSC
ns
Reads
Ge et al
PRISM
E
P. vulgatus reads in lung cancer
Frequency
Genome position
F
CLID
#species/sample
RNA–seq (polyA selected)
RNA–seq (ribosome depletion)
G
CLID
Proportion samples/species
H
CLID
Relative Abundance
I
CLID
Taxa Diversity
J
Dayama Colon
Ghaddar PDAC
Park GC
Priya CRC
CLR in 16S data
–RNA
+RNA
K
TCGA–HNSC
PERMANOVA: R² = 5.8%, p = 0.038
Sequencing center
Baylor College of Medicine
Broad Institute of MIT and Harvard
Harvard Medical School
MD Anderson – Institute for Applied Cancer Science
MD Anderson – RPPA Core Facility (Proteomics)
PCoA1
PCoA2
L
TCGA–HNSC
PERMANOVA: R² = 13%, p = 0.074
IRB center
ABS – IUPUI
Greater Poland Cancer Center
International Genomics Consortium
Johns Hopkins
MD Anderson Cancer Center
Medical College of Georgia
UNC
University Health Network, Toronto
University of Miami
University of Michigan
University of Pittsburgh
Vanderbilt University
M
TCGA–COAD
PERMANOVA: R² = 1.8%, p = 0.25
Sequencing center
Baylor College of Medicine
Harvard Medical School
N
TCGA–COAD
PERMANOVA: R² = 10.5%, p = 0.44
IRB center
BLN – Baylor
Candler
Christiana Healthcare
Greater Poland Cancer Center
Harvard
ILSbio
Indivumed
International Genomics Consortium
MSKCC
UNC
University of Pittsburgh
O
PRISM run time
Microbial read counts
Time (min)

Figure S3. Assessing technical factors that influence microbial capture, Related to Figure 3 and to STAR Methods. (A) Scatter plot compare the number of microbial reads per sample in colon and head and neck cancer in TCGA identified by PRISM vs. by KrakenUniq in Ge et al. Each point represents one sample. **(B)** Scatter plot of the number of taxa per sample identified by PRISM vs. Ge et al. **(C)** Boxplots comparing the proportion of microbial reads assigned to HPV in cervical and head and neck cancer in TCGA by PRISM and Ge et al. ns, no significant difference. **(D)** Boxplots comparing the number of reads assigned to HPV by PRISM and Ge et al. in the same samples. **(E)** Bar plot of the genome position mapped to *Phocaeicola vulgatus* in lung cancer in CPTAC. Nearly all reads mapped to ribosomal RNA. **(F)** Boxplots showing the number of species per sample detected in CLID in poly-A selected (PAS) vs. ribosomal-depletion (RD) RNA-seq. **(G)** Boxplots comparing the proportion of samples in which each taxon in CLID was detected in PAS vs. RD samples. (**H**) Boxplots comparing the relative abundance (proportion of microbial reads to human reads) of contaminants in CLID detected in PAS vs. RD samples. Contaminants are presumed to be present at a similar level of total abundance in these samples. (**I**) Boxplots comparing the diversity of taxon read counts in CLID in PAS vs. RD samples. **(J)** Boxplots of relative abundance (CLR, centered log ratio) in 16S sequencing of species also detected (+RNA) or not detected (-RNA) in RNA-seq performed on the same samples. Boxplots are grouped by study of origin. **(K)** Principal coordinate (PCoA) plot of TCGA head and neck cancers computed based on Aitchison distances. Each point represents a sample. Points are colored by their sequencing center with PERMANOVA testing for sequencing center batch effects. **(L)** PCoA as in (K) testing for IRB institute. **(M)** PCoA as in (K) for TCGA colon cancer. **(N)** PCoA plot as in (L) for TCGA colon cancer. (**O**) Scatter plot of PRISM run time vs. number of microbial reads identified for a subset of samples. Each point is a sample. In all panels, boxplots show median (line), 25^{th} and 75th percentiles (box) and 1.5xIQR (whiskers), and points represent outliers.